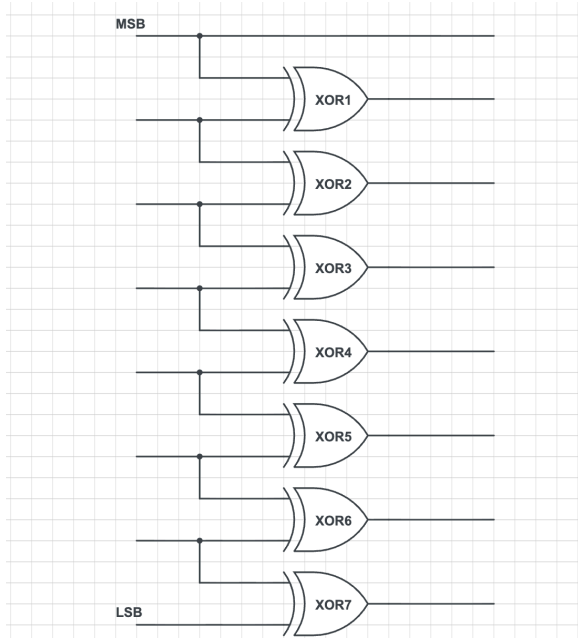
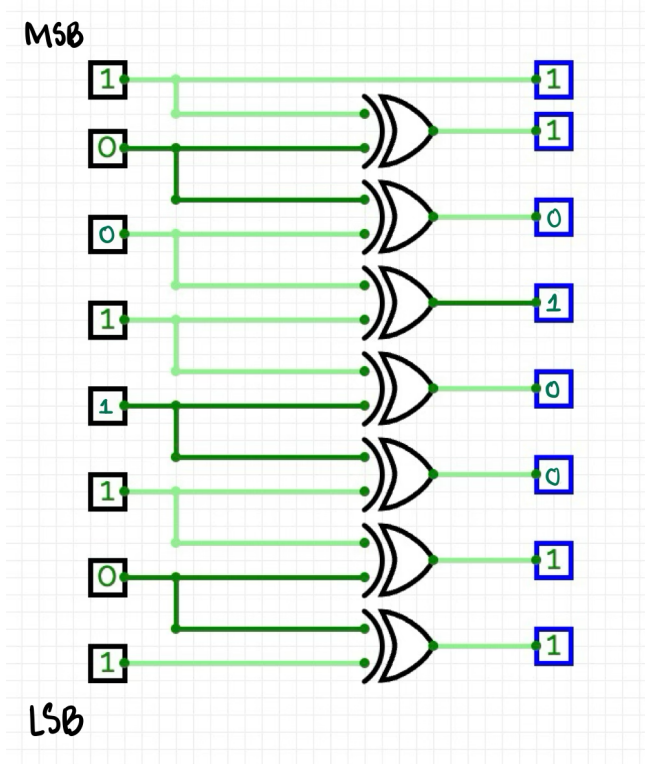


Habib University
Digital Logic and Design - EE/CE-172/222
Assignment 2-Part A Solution Manual
Spring 2026

Q1) a)



b)



c) Binary counting can cause many bits to change simultaneously. Gray encoding guarantees that consecutive values differ in only one bit. During transitions, multiple simultaneous changes can cause errors or unstable signals - causing the system to crash.

Q2) a) $F' = ((X + Y' + Z)(X' + Y + Z')(X' + W))'$
 $= (X + Y' + Z)' + (X' + Y + Z')' + (X' + W)'$
 $= X'Y Z' + X Y' Z + X W'$

[De Morgan's Theorem]

[De Morgan's Theorem, Involution Law]

Truth Table using Original Expression :

W	X	Y	Z	$X+Y'+Z$	$X'+Y+Z'$	$X'+W$	F	F'
0	0	0	0	1	1	1	1	0
0	0	0	1	1	1	1	1	0
0	0	1	0	0	1	1	0	1
0	0	1	1	1	1	1	1	0
0	1	0	0	1	1	0	0	1
0	1	0	1	1	0	0	0	1
0	1	1	0	1	1	0	0	1
0	1	1	1	1	1	0	0	1
1	0	0	0	1	1	1	1	0
1	0	0	1	1	1	1	1	0
1	0	1	0	0	1	1	0	1
1	0	1	1	1	1	1	1	0
1	1	0	0	1	1	1	1	0
1	1	0	1	1	0	1	0	1
1	1	1	0	1	1	1	1	0
1	1	1	1	1	1	1	1	0

Truth Table Using Complimented Expression:

W	X	Y	Z	$x'yz'$	$XY'Z$	XW'	F'
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0
0	0	1	0	1	0	0	1
0	0	1	1	0	0	0	0
0	1	0	0	0	0	1	1
0	1	0	1	0	1	1	1
0	1	1	0	0	0	1	1
0	1	1	1	0	0	1	1
1	0	0	0	0	0	0	0
1	0	0	1	0	0	0	0
1	0	1	0	1	0	0	1
1	0	1	1	0	0	0	0
1	1	0	0	0	0	0	0
1	1	0	1	0	1	0	1
1	1	1	0	0	0	0	0
1	1	1	1	0	0	0	0

- b) i. $A \text{ AND } B = 10100000$
 ii. $A \text{ OR } B = 10111101$
 iii. $A \text{ XOR } B = 00011101$

Q3)

$$\begin{aligned}
 \text{i. } F &= (XY) + (X(Y' + Z)) + (Y(Y + Z)) \\
 &= (XY) + (X(Y' + Z)) + Y && [\text{Absorption Law: } Y(Y + Z) = Y] \\
 &= Y + (XY) + (X(Y' + Z)) && [\text{Commutative Law}] \\
 &= Y + (X(Y' + Z)) && [\text{Absorption Law: } Y + (YX) = Y] \\
 &= Y + (XY') + (XZ) && [\text{Distributive Law}] \\
 &= Y + (Y'X) + (XZ) && [\text{Commutative Law}] \\
 &= (Y + X) + (XZ) && [\text{Distribution + Absorption Law: } Y + Y'X = Y + X] \\
 &= Y + X && [\text{Absorption Law: } X + XZ = X]
 \end{aligned}$$

$$\begin{aligned}
 \text{ii. } F &= (A + B) \cdot (A + C) \cdot (A' + B) \\
 &= (A \cdot B) + (A \cdot C) + (A \cdot A') + (B \cdot C) + (B \cdot A') + (A \cdot B \cdot C) + (A \cdot C \cdot A') + (B \cdot C \cdot A') \\
 &[\text{Distributive Law}] \\
 &= (A \cdot B) + (A \cdot C) + 0 + (B \cdot C) + (B \cdot A') + (A \cdot B \cdot C) + 0 + (B \cdot C \cdot A') \\
 &[\text{Complement Law: } A \cdot A' = 0] \\
 &= (A \cdot B) + (A \cdot C) + (B \cdot C) + (B \cdot A') + (A \cdot B \cdot C) + (B \cdot C \cdot A') [\text{Identity Law: } 0 + X = X] \\
 &= (AB) + (BA') + (ABC) + (AC) + (BC) + (BCA') [\text{Commutative Law}] \\
 &= (AB) + (BA') + (BC) && [\text{Absorption Law: } X + X \cdot Y = X] \\
 &= (AB) + (B(A' + C)) && [\text{Distributive Law: } B \cdot A' + B \cdot C = B \cdot (A' + C)] \\
 &= B(A + C) && [\text{Absorption Law: } B \cdot (A' + C) + A \cdot B = B \cdot (A + C)]
 \end{aligned}$$

$$\begin{aligned}
 \text{iii. } F &= w'xyz + wx'yz + wxy'z + wxyz' \\
 &= yz(w'x + x'w) + wx(y'z + yz') && [\text{Expression is already simplified}]
 \end{aligned}$$

Q4) a)

A	B	C	Z
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

b) Minterms = 3,5,6

Max terms = 0,1,2,4,7

c) SOP = $A'BC + AB'C + ABC'$

POS = $(A+B+C)(A+B+C')(A+B'+C)(A'+B+C)(A'+B'+C')$

Q5)

$$\begin{aligned}
 \text{a) } F' &= ((A'+B)(C'+D')(AB'+CD))' \\
 &= (A'+B)' + (C'+D')' + (AB'+CD)' \\
 &= AB' + CD + (AB')'(CD)' \\
 &= AB' + CD + (A'+B)(C'+D')
 \end{aligned}$$

$$\begin{aligned}
 \text{b) } F &= (x(y'z + x'y) + (x+z'))' \\
 &= (x(y'z + x'y))' ((x+z'))' \\
 &= [x' + (y'z + x'y)'](x+z') \\
 &= [x' + (y+z')(x+y)](x+z') \quad [\text{Further simplification not required}]
 \end{aligned}$$

Q6)

a) POS

$$\begin{aligned}
 F &= (A + B') \cdot (A + B' + C) \cdot (A + C) \\
 &= (A + B') \cdot (A + C) \\
 &= (A \cdot A) + (A \cdot C) + (A \cdot B') + (C \cdot B') \\
 &= A + (A \cdot C) + (A \cdot B') + (C \cdot B') \\
 &= A + (A \cdot B') + (C \cdot B') \\
 &= A + (C \cdot B')
 \end{aligned}$$

[Absorption Law: $x \cdot (x + y) = x$]

[Distributive Law]

[Idempotent Law: $A \cdot A = A$]

[Absorption Law: $A + (A \cdot C) = A$]

[Absorption Law: $A + (A \cdot B') = A$]

b) Non standard / SOP

$$F = (X \cdot Y') + (X \cdot (Y + Z')) + (Y \cdot (X' + Z))$$

$$= X \cdot (Y' + Y + Z') + (Y \cdot (X' + Z))$$

[Distributive Law]

$$= X \cdot ((Y + Y') + Z') + (Y \cdot (X' + Z))$$

[Associative Law]

$$= X \cdot (1 + Z') + (Y \cdot (X' + Z))$$

[Complement Law: $Y + Y' = 1$]

$$= X + (Y \cdot (X' + Z))$$

[Identity Law]

$$= X + (Y \cdot X') + (Y \cdot Z)$$

[Distributive Law]

$$= X + (X' \cdot Y) + (Y \cdot Z)$$

[Commutative Law]

$$= (X + Y) + (Y \cdot Z)$$

[Distribution + Absorption]

$$= X + Y$$

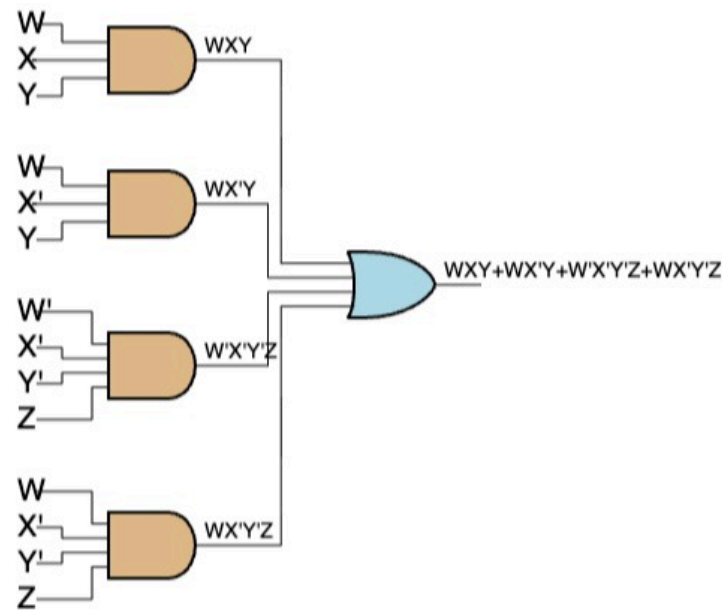
[Absorption Law: $Y + (Y \cdot Z) = Y$]

Q7)

a)

W	X	Y	Z	$WX'Y'Z$	$W'X'Y'Z$	$WX'Y$	WXY	F
0	0	0	0	0	0	0	0	0
0	0	0	1	0	1	0	0	1
0	0	1	0	0	0	0	0	0
0	0	1	1	0	0	0	0	0
0	1	0	0	0	0	0	0	0
0	1	0	1	0	0	0	0	0
0	1	1	0	0	0	0	0	0
0	1	1	1	0	0	0	0	0
1	0	0	0	0	0	0	0	0
1	0	0	1	1	0	0	0	1
1	0	1	0	0	0	1	0	1
1	0	1	1	0	0	1	0	1
1	1	0	0	0	0	0	0	0
1	1	0	1	0	0	0	0	0
1	1	1	0	0	0	0	1	1
1	1	1	1	0	0	0	1	1

b)



c)

$$\begin{aligned}
 F &= (WXY) + (WX'Y) + (W'X'Y'Z) + (WX'Y'Z) \\
 &= (WYX) + (WYX') + (W'X'Y'Z) + (WX'Y'Z) && \text{[Commutative Law]} \\
 &= WY(X + X') + (W' \cdot X' \cdot Y' \cdot Z) + (W \cdot X' \cdot Y' \cdot Z) && \text{[Distributive Law]} \\
 &= WY + (W'X'Y'Z) + (WX'Y'Z) && \text{[Complement Law]} \\
 &= W \cdot Y + (W'X'Y'Z) + (WX'Y'Z) && \text{[Identity Law]} \\
 &= WY + X'Y'ZW' + X'Y'ZW && \text{[Commutative Law]} \\
 &= WY + X'Y'Z(W' + W) && \text{[Distributive Law]} \\
 &= WY + X'Y'Z1 && \text{[Complement Law]} \\
 &= WY + X'Y'Z && \text{[Identity Law]}
 \end{aligned}$$

[Commutative Law]
 [Distributive Law]
 [Complement Law]
 [Identity Law]
 [Commutative Law]
 [Distributive Law]
 [Complement Law]
 [Identity Law]

d)

